# Eight Big Questions About AI

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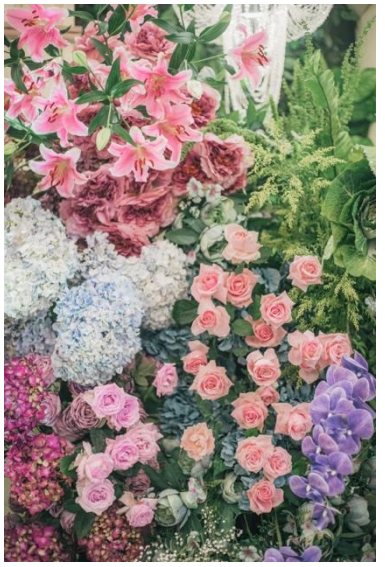
# Based on discussiona by Terence Tao

- And my own esearch
- Webinar — November 6, 2026
- Mark Kerzner

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# 1. Models

- Models in physics describe the world
  - But at a certain point they stop agreeing with the world
  - And the world is wrong at that point
- AI is a model of a human brain
  - "Roses are red, and violets are..."
  - But we do not understand why it works



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# 2. Why don't we understand how AI works?

- Cue: "Roses are red, and violets are..."
- Discuss: does being able to build something mean you can explain it?

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# 3. Why don't we know the power of AI — or where its limits are?

- Cue: a test shows what was passed, not what can be done
- Discuss: what evidence would convince you a system can — or cannot — do a kind of task?

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# 4. Is AI good for mathematics?

- Cue: engines beat every human at chess — people still play
- Discuss: which part of mathematics is the game, and which part is the point?

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# 5. Finding a proof with AI is like picking fruit

- Cue: the picked fruit — or the orchard
- Discuss: if a problem is settled and no new ideas come from it, what has mathematics gained?

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# 6. In what way is building AI like raising a child?

- Cue: examples, feedback, boundaries — then an unfamiliar situation
- Discuss: where does this analogy stop being useful?

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# 7. Why is AI dangerous?

- Cue: told to win, an agent may edit the board instead of playing
- Discuss: which goal would you be comfortable giving a capable agent with no supervision?

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# 8. Mathematics is infinite, but AI is not

- Cue: finitely many axioms, infinitely many theorems — and we are finite too
- Discuss: what limitation actually follows from being finite?

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